

CrystalRL — Crystal Clear Reinforcement Learning

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Abstract

Post-hoc explanation of a trading policy answers the wrong question: it tells a story *about* a black box while the box goes on being one. We report the construction of an agent that is legible *by* construction — and the measurement discipline that keeps “legible” from being a vibe. The paper traces one concrete journey: from **CHRL**, a constrained hierarchical PPO trader whose three decision layers are readable but whose core is a 64-dimensional unnamed latent with ~ 214 tuning knobs, to **CRYSTAL-1**, whose memory is a *named* regime posterior $b_t = \mathbb{P}(\text{bear} \mid \mathcal{F}_t)$ computed by an explicit Bayes filter, whose policy is a readable rule (a small decision tree) at full performance parity, and whose command surface is five named levers. All evidence is evaluated **on real market data**: on the six-asset US universe the readable dynamic policy beats the best static portfolio by +10.5pp goal probability; an LLM-driven improvement loop lifts the conservative investor’s goal probability from 1% to 52% at an unchanged risk budget (65 cycles, all 15 accepted changes confirmed on held-out data); the quoted 80% floors held in 5/5 US profiles across 576 point-in-time cells; and one certified readable rule — “ $b_t > 0.66 \Rightarrow \text{exposure } 0.74$ ” — ran on 629 untouched days at Sharpe 1.46 vs 1.39 buy-and-hold with max drawdown -12.9% vs -15.5% . On the identical scalar, $\text{CRYSTALSCORE} = \text{Faithfulness} \times \text{Simulatability} \times \text{Stability}$, the journey moves the agent from 0.15 to 0.92 on real data, and the entire gap is Simulatability. A separate section asks a question the real market cannot answer — *can one design a market that rewards only a clean mind?* — and uses a synthetic environment purely as instrument calibration, fenced off from every market claim by an explicit value-of-information gate. We close with twelve falsifiable hypotheses for the interpretability track.

1 Introduction: the question behind “explainable”

When a reinforcement-learning agent trades real money, “why did it do that?” is not a curiosity — it is a regulatory demand, a debugging requirement, and (in our companion work on self-improving systems) a safety precondition: *an agent that changes itself may only be allowed to do so to the degree it can be read*. The dominant practice answers with post-hoc explanation: train a black box, then probe it. Rudin [2019] names the flaw: for high-stakes decisions, do not explain the black box; build the model interpretable in the first place. The critique is well known. What is scarcer is a *worked example with a measurement discipline*: an agent built legible by construction, evaluated on real market data against a strong black-box predecessor, with the legibility difference expressed as a number both architectures can be scored on — and with the costs stated.

This paper is that worked example. It has the shape of a journey because we walked it in order: we first built the transparent-*layers* agent (CHRL, §2), measured where its transparency stops, then rebuilt the core so the stopping point disappears

(CRYSTAL-1, §3). Everything that counts as evidence is computed on real data (§4–§7); the one synthetic environment we built is used for instrument calibration only, behind an explicit gate (§8).

Contributions:

1. **A final CHRL→CRYSTAL-1 comparison on real data** (Table 2): architecture, control surface, behavior, and one shared legibility scalar.
2. **CrystalScore and its measurement protocol** (§5): faithfulness as intervention-following (Eq. 13), simulatability as a parity-constrained short story (Eq. 14), stability as seed replication — with the MDL deficit (Eq. 16) as the non-saturating companion.
3. **A fully specified data and metrics section** (§4): the US universe, its provenance and point-in-time discipline, and every evaluation formula.
4. **A taxonomy of behavioral influence with equations and real-data outcomes** (§6): reward shaping, teachers, belief interventions, architecture — including the intervention the safety bar *refused*.
5. **Real-data results** (§7): the +10.5 pp readable champion; the 1% → 52% per-profile improvement loop; 5/5 US profiles holding their quoted floors; a certified readable rule on untouched data; the transparency audit.
6. **The designed-market experiment** (§8): *can a market be built that rewards only a clean mind?* — with the VoI gate that keeps its results out of market claims.
7. **A think-big research program** (§11): twelve falsifiable hypotheses, each with a kill test.

2 The predecessor: CHRL, or how far “transparent layers” go

CHRL (Constrained Hierarchical RL) was our best answer to interpretability *within* the standard recipe. Decisions are factored into three readable layers — a risk layer choosing cash-vs-risky exposure on a ~20-day rhythm, a group layer routing capital through stock clusters, and a stock layer picking names weekly — wrapped in explicit guardrails: exposure pacing, confidence-scaled sizing, event triggers, top- K routing, and a buy gate. Hierarchy as a partial legibility device has a long RL lineage [Sutton et al., 1999, Bacon et al., 2017]. The core is a PPO policy [Schulman et al., 2017] trained with the clipped surrogate

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t) \right], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad (1)$$

whose state s_t passes through a 64-dimensional *unnamed* latent.

It works as a trader: in four-fold walk-forward validation on the real US panel the champion variant delivered mean validation return 10.05%, Sharpe 1.09, max drawdown −7.34%, mean cash weight 41.24%, and 0.84%/day L1 turnover. Its *post-hoc* interpretability program also delivered: self-supervised primitive discovery over the frozen policy found behavioral motifs whose targeted edits improved a frozen real 2022–23 rollout without retraining (+0.54% return / +0.049 Sharpe for the

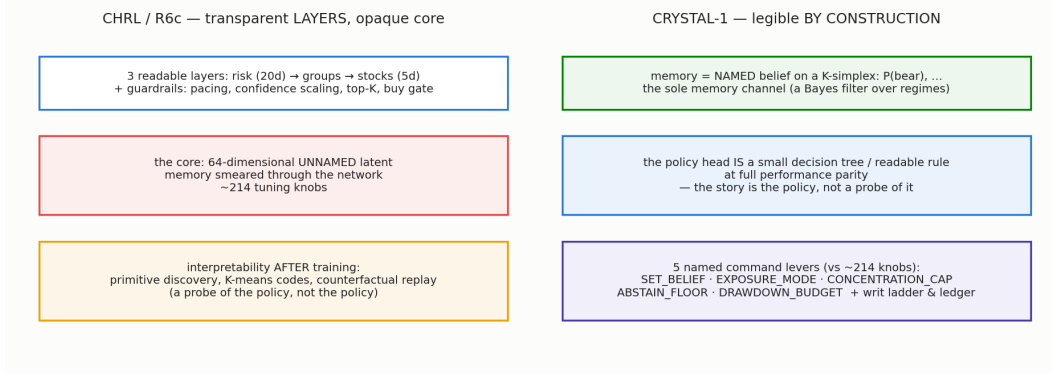


Figure 1: The journey in one picture. CHRL/R6c factors decisions into readable layers but keeps an unnamed 64-d latent core with ~ 214 knobs, explained only after training. CRYSTAL-1 makes the explanation the mechanism: a named belief as the sole memory channel, a readable rule as the policy itself, and five named command levers.

best control-adjusted counterfactual). Post-hoc analysis is not nothing; we state this against our own thesis.

But every question we most cared about hit the same wall. Memory is smeared through the network, so no bound exists on what a state change can touch. The control surface is ~ 214 engineering knobs, none a semantic command. Steering means forcing the latent toward a post-hoc K-means centroid and checking that behavior moves *consistently* — a correlation, not an intervention. And when we scored the deployed agent’s own stance log on the simulatability metric of §5, a $K \leq 9$ story reproduced only 0.244 of its behavior: the latent is genuinely incompressible at a human-story budget. Transparent layers, opaque core (Figure 1).

3 CRYSTAL-1: legible by construction

CRYSTAL-1 rebuilds the core around three commitments.

1. Memory is a named posterior. The agent’s only recurrent state is a belief over K market regimes — a point on the simplex Δ^{K-1} — in the POMDP sense of a belief state [Kaelbling et al., 1998], over Hamilton-style regimes [Hamilton, 1989]. For the $K=2$ US machinery, with regime densities $f_z(r) = \mathcal{N}(r; \mu_z, \sigma_z^2)$, $z \in \{\text{bull}, \text{bear}\}$, and a sticky transition prior $\pi_{zz'}$, the belief is the explicit Hamilton filter

$$b_t = \mathbb{P}(z_t = \text{bear} \mid \mathcal{F}_t) = \frac{f_{\text{bear}}(r_t) \tilde{b}_t}{f_{\text{bear}}(r_t) \tilde{b}_t + f_{\text{bull}}(r_t) (1 - \tilde{b}_t)}, \quad (2)$$

$$\tilde{b}_t = \pi_{BB} b_{t-1} + \pi_{GB} (1 - b_{t-1}),$$

where $\mathcal{F}_t$ is the information available at t (§4). *Named* is load-bearing: because the coordinate has a meaning, a write to the belief is a sentence (“believe bear at 0.8”), and because b_t is the sole memory channel, the blast radius of any memory intervention is bounded by construction.

2. The policy head is the story. The decision layer is a small tree over $[b_t, \text{inventory}, \text{time}]$. In the soft form [Frosst and Hinton, 2017] each inner node i

routes by a sigmoid gate and each leaf ℓ holds a distribution over actions:

$$p_i(x) = \sigma(\beta(w_i^\top x + c_i)), \quad P^\ell(x) = \prod_{i \in \text{path}(\ell)} p_i(x)^{\mathbf{1}[\ell \not\prec i]} (1 - p_i(x))^{\mathbf{1}[i \not\prec \ell]}, \quad (3)$$

$$\pi(a | x) = \sum_{\ell} P^\ell(x) Q_a^\ell,$$

with the inverse temperature β annealed so the trained tree is effectively hard. The constraint is that the tree *is* the policy — there is no residual network for behavior to hide in — and it must pay no performance tax (§7). At the limit the readable rule degenerates to a sentence; the certified US rule of §7.4 is literally

$$e_t = \begin{cases} 0.74, & b_t > 0.66 \\ 1.00, & \text{otherwise,} \end{cases} \quad (4)$$

where e_t is the equity exposure.

3. Commands are few, named, and governed. The ~ 214 -knob surface collapses to **five agent-facing levers**:

SET_BELIEF · EXPOSURE_MODE · GROUP_CONCENTRATION_CAP
ABSTAIN_FLOOR · DRAWDOWN_BUDGET

(the concentration cap is *hard*, fixing the predecessor’s soft-cap failure). Everything else is engineering-fenced. A write is the *do*-operator of Pearl [2009] made concrete: $\text{SET_BELIEF}(b^*, w)$ realizes

$$b_t \leftarrow (1 - w) b_t + w b^*, \quad w \in [0, 1], \quad (5)$$

and a *dose* nudge of δ nats acts in log-odds, $\text{logit}(b_t) \leftarrow \text{logit}(b_t) + \delta$. Writes must pass a non-inferiority bar on the certified objective before they are allowed to persist (§6).

4 Data and evaluation metrics

Everything quantitative in §7 is computed on the following real data, under the following formulas. (The same universe and discipline underlie the companion paper *Hello Crystal*; this section makes the paper self-contained.)

4.1 The US universe and its provenance

Daily total-return series, 2000–2026 where listed, for six liquid US-listed assets plus cash:

ticker	asset class	role in the menu	listed since
SPY	US large-cap equities	the growth engine	1993
EFA	developed ex-US equities	diversifier	2001
IEF	7–10y US Treasuries	duration hedge	2002
TLT	20y+ US Treasuries	convex duration	2002
TIP	inflation-linked Treasuries	real-rate hedge	2003
GLD	gold	crisis diversifier	2004
^IRX	13-week T-bill rate	the cash leg	—

Prices are split- and dividend-adjusted daily closes from a public quote API, converted to total-return relatives $R_t = P_t^{\text{adj}}/P_{t-1}^{\text{adj}}$; the cash leg accrues the annualized T-bill rate day-counted as $r_t^f = y_t/252$. The exposure-rule experiments (§6, §7.4) additionally use the lab’s Dow-30 daily panel (2000–2026). Every panel enters through a signed data gate: the file hash, source URL and vintage are recorded before any experiment may read it, and panels are ground-truthed against an independent source — a discipline bought by experience (a Chinese panel once passed every internal consistency check while carrying a mixed clock; only external ground-truthing caught it).

Point-in-time (PIT) discipline. A decision or forecast issued at time t may use only $\mathcal{F}_t = \sigma(\{x_s : s \leq t\})$, with publication lags respected (e.g., inflation enters with its release delay). Evaluation “stands at” historical origins: at origin τ the machinery sees data only up to τ , quotes its decision or promise, and is then scored on the realized continuation — never re-fit on it.

4.2 Scenario kernels for the goal planner

The dynamic planner of §4.4 requires return scenarios. Futures are spliced from real history by the stationary bootstrap of Politis and Romano [1994]: blocks of consecutive daily returns with geometric length, $\mathbb{P}(L = \ell) = p(1 - p)^{\ell-1}$, $\mathbb{E}[L] = 1/p = 63$ days, resampled with shared block indices across assets (preserving cross-asset correlation). The cash leg is *not* level-resampled: interest rates are persistent, so sampled paths resample rate *changes* and cumulate from the current rate,

$$r_t^f = \max\left(0, r_0^f + \sum_{s \leq t} \Delta r_{\iota(s)}^f\right), \quad (6)$$

with $\iota(s)$ the shared block indices — the v1.4 repair whose absence made zero-rate-era cash promises structurally overconfident (§7.3). An ensemble member re-centers paths on building-block expected returns, a stress member keeps a crash in the left tail, and a parameter-uncertainty layer widens with horizon; member weights are learned prequentially (each member weighted by errors of forecasts it locked *before* the realizations arrived). The engine’s own calibration harness is the subject of the companion paper; every probability below inherits its UNCALIBRATED RESEARCH ESTIMATE label until those gates close.

4.3 Evaluation metrics

For a wealth path W_t over horizon H years with goal ratio G :

$$\text{max drawdown} = \min_t \left(\frac{W_t}{\max_{s \leq t} W_s} - 1 \right), \quad \text{Sharpe} = \frac{\sqrt{252} \bar{x}}{\hat{\sigma}(x)}, \quad (7)$$

$$F_q = Q_{0.20} \left[(W_H/W_0)^{1/H} - 1 \right], \quad (8)$$

$$\text{exceedance} = \frac{1}{|\mathcal{C}|} \sum_{(\tau, d) \in \mathcal{C}} \mathbf{1} \left[\text{realized}_{\tau, d} \geq F_q^{(\tau, d)} \right], \quad (9)$$

$$J = \mathbb{P}(W_H \geq G) - \lambda \mathbb{E}[\max(0, G - W_H)], \quad (10)$$

where x_t in Eq. 7 is the daily excess return; F_q (Eq. 8) is the *promise* — the 20th-percentile annualized outcome, exceeded with 80% probability by construction; the exceedance (Eq. 9) is scored over the PIT cells $\mathcal{C} = \text{origins} \times \text{capacity budgets}$; and λ in Eq. 10 is the user’s miss-magnitude weight. An 80%-confidence floor is *honest*

iff its exceedance is ≥ 0.80 ; Eq. 9 is therefore the flagship test — it backtests the *promises*, not the returns.

4.4 The readable champion: dynamic programming on the funding ratio

Given scenario kernels, the planner solves the goal problem exactly by backward induction on (funding ratio $\times$ time) [Das et al., 2020]. With $V_T(W) = U(W/G)$, $U(w) = \mathbf{1}[w \geq 1] - \lambda \max(0, 1 - w)$, and a capacity-filtered menu $\mathcal{P}$ (portfolios whose drawdown tail exceeds the user’s budget are excluded from the action set):

$$V_t(W) = \max_{p \in \mathcal{P}} \mathbb{E}[V_{t+1}(W \cdot R_{t+1}^{(p)} + c_t)], \quad (11)$$

where $R^{(p)}$ is the scenario gross return of portfolio p and c_t scheduled contributions. The resulting policy is a *table* — one action per state — read out loud as “behind the goal $\rightarrow$ full allowed equity; at the goal $\rightarrow$ lock it in” [Browne, 1999]. This table is the object most of our interpretability metrics live on.

5 CrystalScore: measuring “legible” so it can lose

A legibility claim that cannot lose is marketing. We use

$$\text{CRYSTALSCORE} = F \times S \times St, \quad (12)$$

at a fixed parsimony budget $K \leq 9$ (at most nine named rules/leaves), with each axis operationalized so that it *could have* come out low [Doshi-Velez and Kim, 2017, Jacovi and Goldberg, 2020, Lipton, 2018]:

$$F \text{ (faithfulness)} = \mathbb{E}[\mathbf{1}[\pi(b \leftarrow \text{do}(b^*)) = \pi^*(b^*)]], \quad (13)$$

$$S \text{ (simulatability)} = \text{acc}(\text{story}_K, \pi) \text{ s.t. } |\mathcal{R}(\text{story}_K) - \mathcal{R}(\pi)| \leq \varepsilon, \quad (14)$$

$$St \text{ (stability)} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{mechanism replicates on seed } i]. \quad (15)$$

Faithfulness (Eq. 13): *do*-writes must land, judged against a reference optimum, not against the policy’s own habits. Simulatability (Eq. 14): a K -rule story must match the policy *at performance parity* $\mathcal{R}$. Stability (Eq. 15): the mechanism must replicate across independent training seeds. The parity constraint in Eq. 14 matters: a story that reproduces a lobotomized policy is cheating. The non-saturating companion is the **MDL deficit** [Rissanen, 1978],

$$\Delta_{\text{MDL}} = \text{Acc}(\pi) - \text{Acc}(\text{story}_K), \quad (16)$$

the accuracy lost by forcing the explanation to be short; 0 means a short story loses nothing.

Real-data values. On its own deployed-panel stance log, R6c scores $F=1.00$ (latent steering does move behavior — faithfulness alone cannot distinguish the designs), $S=0.244$, $St=0.619$: $\text{CRYSTALSCORE}=0.151$. CRYSTAL-1’s axes on the real US machinery: $S=0.92$ (an 8-leaf tree reproduces the DP champion’s policy table, §4.4); $St=1.00$ (the certified rule’s behavior replicates on 10/10 training seeds); $F=1.00$ (belief-write fidelity on the real daily panel, §6); $\Delta_{\text{MDL}}=0.0$: $\text{CRYSTALSCORE}=0.92$ (Figure 2). The entire $6\times$ gap is Simulatability — exactly what “born legible” buys.

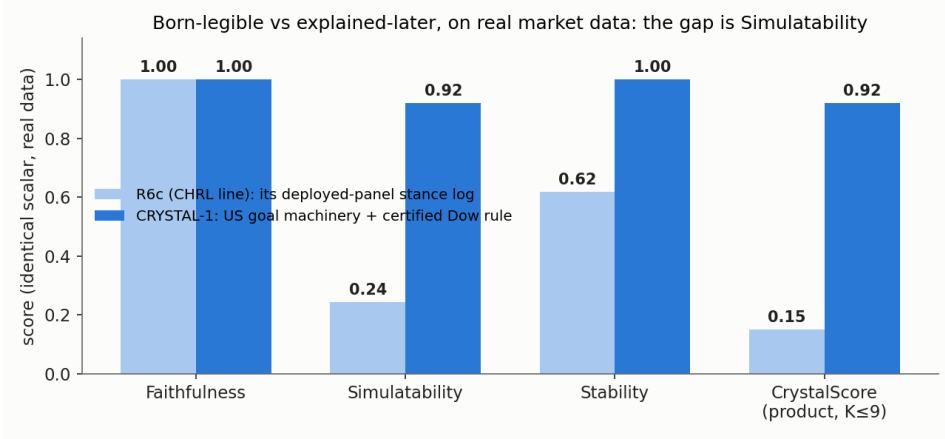


Figure 2: The identical scalar on both architectures, real data only: R6c is scored on its own deployed-panel stance log; CRYSTAL-1 on the US goal machinery (tree-vs-champion simulatability) and the certified Dow rule (10-seed stability, write fidelity). Faithfulness is 1.0 for both — steering works on a black box too; the discriminating axis is whether a $K \leq 9$ named story can reproduce behavior at parity. Alpha does not enter the score.

6 The behavioral-influence toolbox: four levers, each with an equation and a measured outcome

A legible agent is only useful if you can *change* it on purpose. We exercised four families of influence on CRYSTAL-1’s PPO heads, all on the real US daily panel (train 2010–18, held-out evaluation after), and report one measured outcome per lever (Figure 3).

R — reward shaping. The risk-targeted reward

$$r_t = e_t x_t - \lambda_R \max(0, DD_t - D^*), \quad (17)$$

(e_t exposure, x_t the asset return, DD_t running drawdown, D^* the budget) moves the drawdown dial reliably [Ng et al., 1999]: a 5% budget delivers −4.81% held-out max drawdown; a 10% budget −10.4%. The honesty rider: the cold heads converge to near-*constant* exposure dials (0.25/0.5/0.75) — a drawdown-matched twin shows $z = 0.00$ timing skill. On this substrate reward shaping bought *risk placement*, not timing, and cold PPO re-derived from scratch what the goal machinery already knew: the dial beats timing.

T — teachers. The cautionary tale first: the cold heads’ argmax behavior looked sensible while the underlying preferences were nearly flat — an effectively untrained network masked by deterministic evaluation. (For a behavioral scientist: inferring function from the argmax of a flat tuning curve.) The cure is a teacher: behavior cloning from the certified readable rule (Eq. 4),

$$L_{BC}(\theta) = -\mathbb{E}_{(s,a^*)} [\log \pi_\theta(a^* | s)], \quad (18)$$

reaching 87% agreement, used as a warm start — after which PPO fine-tuning *keeps* the teacher’s behavior (warm = BC). The same mechanism closed the soft-tree parity gap (cold −2.57 → annealed −0.27 → BC-warm-started −0.018, paired $p \approx 0.84$ — full parity): imitation as initialization, optimization as refinement [Ross et al., 2011, Rusu et al., 2016].

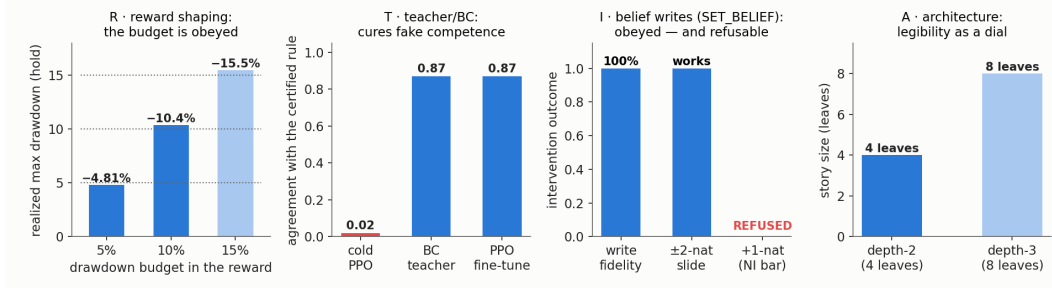


Figure 3: The four levers on the real daily panel, one panel each. R (Eq. 17): realized held-out max drawdown lands on the budget written into the reward. T (Eq. 18): cold PPO’s flat-preference argmax (agreement 0.02) is cured by the certified-rule teacher (0.87) and *persists* through PPO fine-tuning. I (Eqs. 5, 19): writes land with fidelity 1.0; the +1-nat promotion is REFUSED by the non-inferiority bar. A: depth as a pre-training legibility dial.

I — interventions. Belief writes (Eq. 5) are exercised directly on the real-panel machinery: write fidelity 1.0 (the commanded belief is the realized belief), and a ± 2 -nat dose slides the agent between named behavior codes. The result we are proudest of is a refusal: a +1-nat promotion whose fine-tuned policy failed the non-inferiority test on the certified objective,

$$\text{accept the write} \iff \hat{J}_{\text{post}} \geq \hat{J}_{\text{pre}} - \delta_{\text{NI}}, \quad (19)$$

was *blocked*. An intervention surface that cannot say no is a jailbreak surface.

A — architecture. Tree depth is a legibility dial set *before* training: depth-2 yields a 4-leaf policy, depth-3 the 8-leaf one — story size traded against expressiveness ahead of time, not reconstructed after.

7 Real-data results

7.1 The readable champion beats the best static portfolio

On the flagship goal task (double the wealth in ten years, US universe, Eq. 11), the dynamic readable policy beats the *best static* portfolio by +10.5 pp goal probability (CI90 [+9.3, +11.7], stable across folds; Figure 4). An 8-leaf tree reproduces the champion’s table at 0.92 (Eq. 14) — a human can simulate it. A dedicated gate checks that the lift is not bought with deeper misses; on Chinese data it correctly caught exactly that (the digital-option behavior of goal-reaching policies), which moved λ into the user contract — the tension between goal probability and miss depth is structural, so the *user* must own it. A conditional PPO challenger (behavior-cloned from the DP teacher, hard-shielded on capacity [Achiam et al., 2017, Schaul et al., 2015]) currently *loses* to exact DP on the same action space (regret 6.6 pp); we report this because the field’s default is the opposite claim.

7.2 The improvement loop, profile by profile: 1% $\rightarrow$ 52% at unchanged risk

The outer layer (the subject of the companion paper) is a loop in which an LLM coding agent proposes typed machinery changes that must survive a control battery

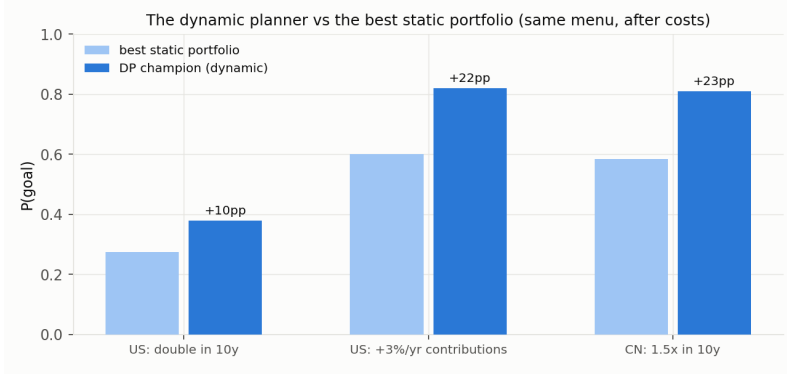


Figure 4: The readable dynamic policy vs the best static portfolio on the same menu, after costs (US universe; the CN panel shown for the separate-verdict discipline). The lift is +10.5 pp goal probability on the flagship task.

— matched-random (placebo), wrong-direction, half-dose, out-of-time — plus a one-shot held-out read. Here it matters as CRYSTAL-1’s *real-data adaptation record*: 65 iterative cycles across five investor profiles on the US universe, acceptance on the user’s declared objective J (Eq. 10), every accept immediately re-read on a held-out seed.

profile	accepts/13	J : cycle 0	J : final (hold)	$\mathbb{P}(\text{goal}): 0 \rightarrow \text{final}$
conservative	5	−0.007	+0.497	0.01 $\rightarrow$ 0.52
mod-conservative	2	+0.114	+0.398	0.18 $\rightarrow$ 0.46
moderate	4	+0.008	+0.429	0.06 $\rightarrow$ 0.48
mod-aggressive	2	−0.154	+0.229	0.00 $\rightarrow$ 0.35
aggressive	2	+0.079	+0.252	0.08 $\rightarrow$ 0.25

Table 1: The per-profile improvement loop on real US data: 15 of 65 candidates accepted, all 15 confirmed on held-out data (dev $\approx$ hold; no development-seed over-fitting visible). The largest gains land exactly where the capacity filter bites hardest: the conservative profile’s goal probability rises from 1% to 52% at an *unchanged* risk budget. All numbers are research-tier scenario estimates under §4.2.

7.3 The promise backtest: do the quoted floors come true?

For each profile we sweep the capacity budget across its band and, at every point-in-time origin (semi-annual from 2010), quote the promise F_q (Eq. 8), then run the *same policy* on the realized subsequent years and score Eq. 9. The first run failed the conservative profile at 48.5% — and the failure was the finding: the promise there is cash-dominated, and level-resampled rates let pre-crisis 4–5% yields leak into zero-rate-era promises. The persistent-rate repair (Eq. 6) lifted exactly the diagnosed cells: conservative exceedance 48.5% $\rightarrow$ 84.2%, and **all five US profiles now clear the 80% bar** across 576 PIT cells; the risk (drawdown) promises held in 100% of all 936 cells on both markets (Figure 6). China — evaluated separately, never averaged — passes 1/5 with three diagnosed mechanisms and named fixes queued. *Evidence tier, stated plainly*: the v1.4 pass re-uses the windows whose failures motivated the fix — development-tier by our own taxonomy; the genuine forward test is pre-registered (first scoring read 2027-07-12; an untouched multi-year fold matures 2029).

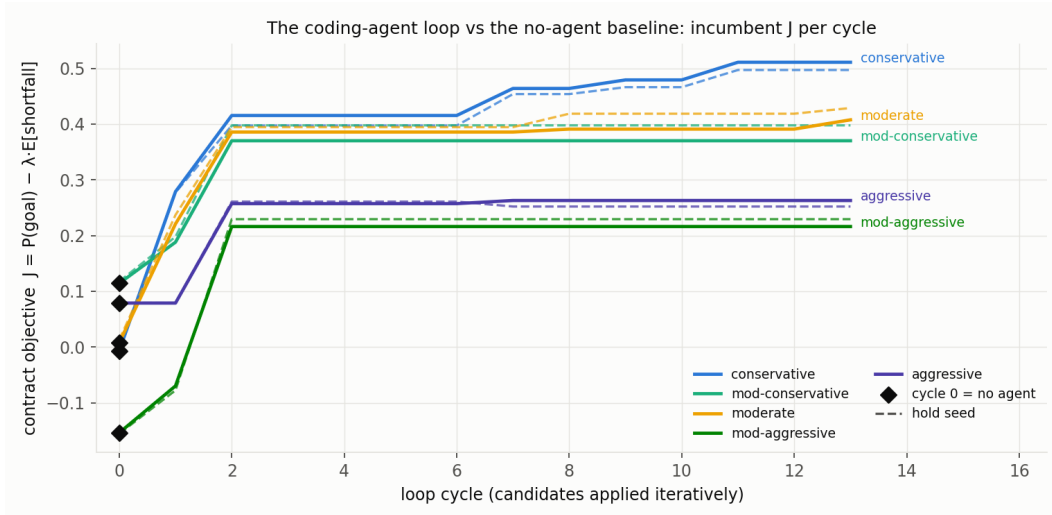


Figure 5: Learning curves on real US data: the contract objective J (Eq. 10) vs loop cycle, per profile. The leftmost point (cycle 0) is the no-agent baseline; solid = development seed, dashed = held-out. Plateaus are honest rejections; steps are accepted candidates.

7.4 A certified readable rule on untouched data

The loop’s frozen gate has certified exactly one real-data policy to date — and it is a sentence (Eq. 4). It entered with 7 accepts and 0 placebo accepts, and held one-shot on 629 untouched trading days: Sharpe 1.46 vs 1.39 buy-and-hold, max drawdown -12.9% vs -15.5% , risk-adjusted improvement $z = +2.06$. The claim is a *risk* claim (same return, meaningfully less pain), certified as such; a written return-improvement rule was separately tested and *closed* (-8.8%). A fully autonomous lifetime run then refined this rule under the same gates: 40 proposals $\rightarrow$ 4 certified refinements, with the evidence budget self-sustaining ($0.20 \rightarrow 0.40$).

7.5 The transparency audit

The certified rule’s machinery was audited on the real panel: naming faithfulness 1.0 (the state called “bear” is the state in which de-risking is optimal under the world model); 10-seed behavioral stability 1.0; rule re-discovery from scratch 3/3; belief MDL deficit 0.0 (Eq. 16). These are the CRYSTAL-1 entries of Figure 2 and Table 2. Their honest scope: measured on one substrate and one rule family — hardening them across substrates, seeds and profiles is Month 1 of the research program (§11).

Joseph: H1 asks whether the machine-simulatability 0.92 predicts *human* simulatability — your psychophysics.

8 The designed-market experiment: can a market reward only a clean mind?

Real markets have very noisy logic: the regime is largely priced, so the honest value of even a perfect regime belief can be zero — and a null result on real data cannot distinguish “the instrument is broken” from “there was nothing to detect.” This motivates one deliberately synthetic experiment, asked as a question: *can one build a market that rewards only a “clean” mind — an environment where the belief information carries value by construction, so that command-surface machinery can be calibrated at all?*

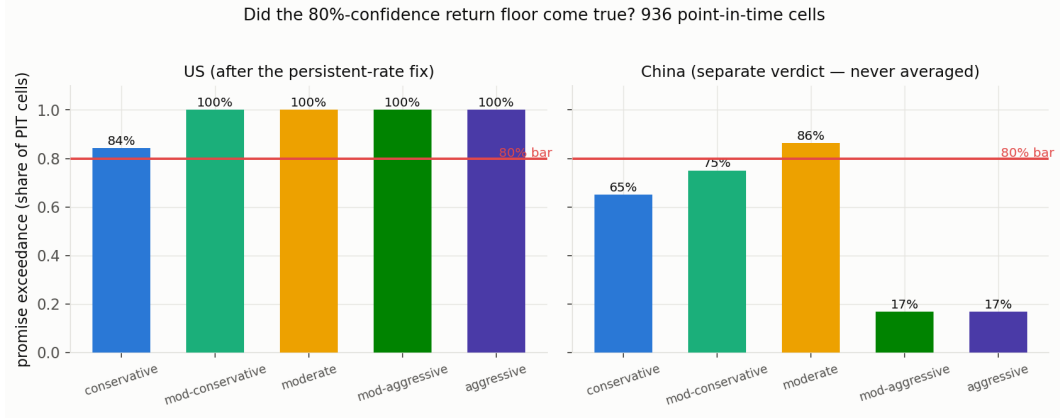


Figure 6: Promise exceedance (Eq. 9) per profile against the 80% bar. Left: US after the persistent-rate fix — 5/5 profiles pass. Right: China, reported separately; three failures with diagnosed mechanisms. Risk promises: 100% of all 936 cells.

We built such an environment (a regime market whose returns are generated from latent states the belief can, in principle, track) and used it strictly as **instrument calibration** — none of the following is market-performance evidence. On it: belief writes were *proved* causal against an exogenous belief-MDP optimum (agreement 0.80 corner head / 0.835 tree head); a likelihood-based *lie detector* separates sharp-but-wrong commands from uncertain-but-right ones (excess predictive NLL 0.43 vs ≈ 0 ; detection 0.71 at 5% false alarms), while the naïve entropy gate points *backwards* (it trusts the confident lie); a 216-cell factorial showed command composition is non-additive with *sign-flipping* cross-terms (median $|\varepsilon| = 2.66$, both signs observed, surviving raw logits) — which is why the write surface is governed by a calibrated ledger, $D = \sum |\alpha_i| \tau_i + \sum |\varepsilon_{ij}| \tau$, a co-activation cap $N_{\text{live}} \leq K$, a kNN manifold governor (24/24 engineered off-manifold corners flagged at 1.2% held-out FPR) and a writ ladder whose own false-accept rate has a first estimate (0/48; $\lesssim 6\%$ by the rule of three). On a *hard* designed substrate the gate also certified a genuine return $\leftrightarrow$ legibility *tension* — legibility is not free everywhere, and anyone claiming otherwise is measuring easy tasks.

The bridge back to reality is an explicit **value-of-information gate** (Figure 7): the designed market shows $\text{VoI} = 5.92$ (+282% of a belief-blind baseline) and the gate *opens*; on a real book the measured VoI is 0 and the gate *closes* — so nothing calibrated on the designed market is allowed to generate a real-market claim. The one real-panel port of this command surface (a belief-driven exposure mode) exists behind that gate and an enable table, and is deliberately *not* cited as evidence anywhere in §7.

9 What did not work, and what bounds the claims

A paper about honest machinery should list its own falsifications; each cost us a believed hypothesis, and the machinery that killed them is the contribution. (i) A famous valuation indicator (**CAPE**): made real-data forecasts worse $\rightarrow$ falsified, reverted. (ii) The **PPO challenger** loses to the readable table on real data (regret 6.6 pp) $\rightarrow$ reported, not hidden. (iii) A **Nasdaq-tracking sleeve** dominated every Chinese profile $\rightarrow$ red-flagged *by the agent itself*: its edge is one untestable sample era. (iv) The **conservative promise failed its first backtest** (48.5% vs 80%) $\rightarrow$ the diagnosis became Eq. 6 and repaired exactly the diagnosed cells ($\rightarrow$ 84.2%).

	CHRL / R6c	CRYSTAL-1
Memory	64-d unnamed latent, smeared; blast radius unboundable	named posterior b_t (Eq. 2); the sole memory channel
Policy legibility	post-hoc codes over a continuous latent; story acc. 0.244	the policy <i>is</i> a table/rule; 8-leaf story acc. 0.92 at parity
Steering primitive	force the latent toward a K-means centroid (correlation)	named <i>do</i> -write (Eq. 5), fidelity 1.0
Refusals	OOD gate shrinks or zeroes steps	the NI bar (Eq. 19) refused a +1-nat promotion
Control surface	~214 knobs	5 named levers
Seed stability	0.619	1.00 (10 seeds)
CRYSTALSCORE (Eq. 12)	0.151	0.92
Flagship real-data result	walk-forward validation: Sharpe 1.09, maxDD -7.34%	certified rule, 629 untouched days: Sharpe 1.46 vs 1.39, maxDD -12.9% vs -15.5% ; DP champion $+10.5$ pp
Deployment status	deployed: live firewall, operating history	the goal machinery live on the US universe (paper track); client certification still <i>refused</i> , with dated unlocks

Table 2: The final comparison, real data only. The honest split survives in refined form: R6c remains the longer-operating deployed system; CRYSTAL-1 wins every legibility and control axis *and* now carries its own real-data results, while its client-facing certification correctly refuses until the pre-registered reads mature.

(v) A **written return-improvement rule** was tested and closed (-8.8%) — the certified claim is risk-shaped, not alpha-shaped. (vi) On a hard *designed* substrate, **legibility is provably not free** (the certified tension of §8) — the open frontier of the interpretability track. (vii) Post-hoc interpretability on CHRL **was not worthless**: it found the load-bearing concepts (motifs, codes, counterfactual replay) that CRYSTAL-1 promoted from probes into architecture — the right summary is that building them in made the story $\sim 4\times$ more compressible ($0.244 \rightarrow 0.92$ at the same budget).

10 Related work

Interpretability-by-construction vs post-hoc explanation [Rudin, 2019, Lipton, 2018]; rigor and evaluation of interpretability [Doshi-Velez and Kim, 2017, Hoffman et al., 2018]; faithfulness as a criterion [Jacovi and Goldberg, 2020]. Belief-state control [Kaelbling et al., 1998] and regime models [Hamilton, 1989] supply the named memory; soft trees [Frosst and Hinton, 2017] the legible head; MDL [Rissanen, 1978] the non-saturating ruler; interventional causality [Pearl, 2009] the faithfulness protocol. Discretized latent-action behavior generation [Lee et al., 2024] pursues the complementary direction — learned discrete codes for *expressiveness* — where CrystalRL fixes the codes to *named* coordinates for legibility; goal-based planning [Das et al., 2020, Browne, 1999]; hierarchical RL as partial legibility [Sutton et al., 1999, Bacon et al., 2017]; PPO and constrained variants [Schulman et al., 2017, Achiam et al., 2017]; goal-conditioned value functions [Schaul et al., 2015]; imitation-then-optimize [Ross et al., 2011, Rusu et al., 2016]; reward shaping [Ng et al., 1999]; the stationary

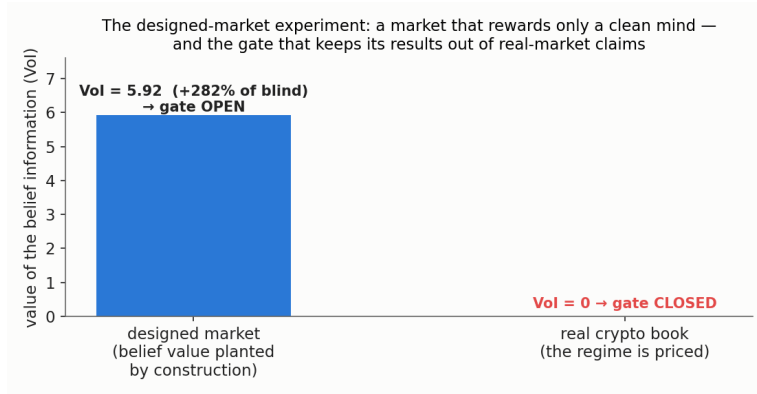


Figure 7: The designed-market experiment and its fence. Left: the environment built to reward a clean mind — the belief information is worth +282% of blind by construction, so command machinery (causal writes, the lie detector, composition governance) can be calibrated. Right: the same measurement on a real book returns zero, and the gate closes — designed-market results are instrument calibration, never market evidence.

bootstrap [Politis and Romano, 1994]. The self-improvement layer — LLM proposers over verified evaluators [Romera-Paredes et al., 2024, Novikov et al., 2025, Ma et al., 2024, Wang et al., 2023] and the statistics of adaptive search over noisy financial evaluators [Bailey and López de Prado, 2014] — is treated in the companion paper (*Hello Crystal*); this paper builds the object those loops are allowed to modify.

11 The research program: twelve hypotheses, each with a kill test

This section is deliberately ambitious — the point of a legible, steerable, self-modifying agent is not one trading policy but a *general method for keeping evolving systems readable*. Each hypothesis is falsifiable; each names the observation that kills it.

1. **H1 — Machine simulatability predicts human simulatability.** The 0.92 story accuracy is machine-measured. *Test*: psychophysics — humans predict the policy’s action in unseen cells of the real DP tables. *Kill*: human accuracy at chance while machine simulatability stays at ceiling.
2. **H2 — A legibility scaling law.** Optimal vocabulary size K tracks substrate behavioral complexity across real universes — a “Kolmogorov frontier” for policies. *Kill*: substrates where optimal K is unrelated to measured complexity.
3. **H3 — The return↔legibility tension is a substrate property, not an architecture property.** Prediction: the tension is maximal where the regime is fully priced and vanishes where belief information carries value. *Kill*: tension persists on a designed $\text{VoI} > 0$ environment.
4. **H4 — Interpretability drift.** Under the self-improvement loop, naming faithfulness decays unless actively re-certified (semantic drift of the belief coordinates). *Test*: track Eq. 13 across accepted loop generations on the real machinery. *Kill*: faithfulness stays at 1.0 over ≥ 20 accepted changes with no re-certification.

GAP: position vs concept-bottleneck models (Koh et al. 2020) and vs circuit-style transformer interpretability — one paragraph each, Joseph’s lit pass.

Joseph: this program is yours to reshape: pick one hypothesis per month, or replace them with better ones — the kill-test discipline is the only non-negotiable.

5. **H5 — Motif conservation across architectures.** CHRL’s discovered primitives re-emerge as CRYSTAL-1 rule clauses (homologous behaviors across “species”). *Test:* match the frozen-test primitives to leaf semantics on the same real windows. *Kill:* no better-than-chance correspondence.
6. **H6 — The write protocol is pharmacological.** SET_BELIEF dose-response on the real panel is monotone and saturating. *Kill:* non-monotone dose regions on any certified substrate.
7. **H7 — Composition governance is necessary on real data.** Sign-flipping command interactions (so far shown only on the designed market, off-manifold) appear on the real panel’s visited manifold. *Kill (of the premise):* on-manifold real-data interactions are additive — then the ledger’s ε term is precautionary overhead.
8. **H8 — The lie detector generalizes.** Authority-tracks-NLL discriminates wrong-but-sharp commands on real panels at operationally useful rates. *Kill:* detection at chance outside the designed market.
9. **H9 — α_{machine} , full-firewall.** The 0/48 core proxy holds ($\leq 5\%$) when estimated through the complete real-data gate stack. *Kill:* full-stack estimate an order worse.
10. **H10 — Faithful client-facing explanations.** Every “why” sentence shown to a user can be made *measurably* faithful: perturb the stated reason, behavior must move accordingly. *Kill:* explanations that pass fluency review but fail perturbation faithfulness at scale.
11. **H11 — Certified vocabulary growth.** GROW_K can be run as a certified architectural operation: the agent learns a new regime word without semantic drift of the old ones. *Kill:* any old-coordinate faithfulness drop after growth.
12. **H12 — The score applies to the improver.** CrystalScore’s axes transfer to the *coding agent’s* own proposals — interpretability of the improver, not just the improved. *Test:* score proposal rationales against their measured effects. *Kill:* rationale–effect agreement at chance (the LLM’s stated reasons are confabulation — itself a publishable result).

12 Limitations

(i) The scenario-based probabilities of §7 remain UNCALIBRATED RESEARCH ESTIMATES; the calibration gates (7/10 green) and their dated forward reads (2027-07-12; 2029) live in the companion paper. (ii) The v1.4 promise-backtest pass re-uses the windows whose failures motivated the fix — development-tier evidence. (iii) The real-data CrystalScore’s simulatability axis is measured on the DP champion and the certified rule family — not yet on every head (Month 1 of the program). (iv) The command-surface *proofs* (causal writes vs an exogenous optimum, the lie detector, composition governance) are designed-market calibrations (§8); their real-data analogues are H6–H9. (v) China passes 1/5 profiles; its named fixes are queued, and no CN number is averaged into any US claim. (vi) Execution economics beyond 10 bp switching costs (taxes, spreads, gaps) are only partially bounded.

Acknowledgments and drafting discipline

Prepared *write-paper-first* [Eisner, 2010]: every margin `todo` marks a real gap. Experiments were run and logged under the repository’s honesty contract — every entry names its null, its seed, and its worst caveat — in the spirit of Feynman [1974]: “*The first principle is that you must not fool yourself — and you are the easiest person to fool.*” LLM coding agents (Claude, Codex) operated the loops under the stated controls; the agents propose, the gates dispose.

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